

REPORT APPENDIX AND LOG: THE MALAYSIA-INDIA ESG SYNERGY BLUEPRINT

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This appendix documents the full development process, underlying methodologies, and supporting evidence of the report—ensuring transparency, reproducibility, and traceability of all analytical outputs.

Table of Contents

Appendix Reading Guide	01 - 02
PART I — Development Phases & Process Log	03
Development Phases Timeline	04
Phase 1: Vision & Research Initiation (January 2026)	05
Phase 2: The Strategic Pivot (February 2026)	06
Phase 3: Data Engineering & Integrity (Technical Deep-Dive)	07
Phase 4: Strategic Localization & Critical Thinking (Strategy Development)	08
Phase 5: Digital Architecture & UX Design (MARCH 2026)	09
Phase 6: Model Validation & System Optimization (Final Refinement Phase)	10 - 11
Key Learnings & Reflections	12
PART II — Model Methodology & Validation	13
2.2A Emission Reality Model Methodology and Structural Interpretation	14 - 15
2.2B — Adoption Impact System Methodology	16 - 17
Adoption Curve Model — Methodology	18 - 19
Reduction Contribution Model — Methodology	20 - 22
Emission Trajectory Model — Methodology	23 - 24
PART III — Evidence and Traceability	25 -26
Closing Statement	27
About the Author	28

Appendix Reading Guide

Purpose of this Appendix

This appendix is designed to complement the main report by providing transparency, technical validation, and traceability.

It serves three core functions:

Process Transparency (Phases & Log)

Documents the full development journey, including key decisions, challenges encountered, and how they were resolved.

Methodological Validation (Models)

Provides the mathematical structures and logical frameworks behind the data models used in the report, enabling verification and replication.

Evidence & Traceability

Ensures that all analysis and outputs can be traced back to original sources, tools, and documented processes.

How to Navigate This Appendix

This appendix is structured into three main parts.

Readers may approach it based on their objective:

For Understanding the Development Process

Start with **Part I — Development Phases & Process Log**

⇒ Provides a chronological view of how the report evolved from initial concept to final system design.

For Verifying Data Models

Go to **Part II — Model Methodology**

⇒ Explains the mathematical logic behind:

- Emission Reality Model (2.2A)
- Adoption Impact System (2.2B)

This section is intended for readers who want to validate assumptions or replicate the modeling framework.

For Evidence and Validation

Refer to **Part III — Evidence & Traceability**

⇒ Contains links to:

- Source datasets (HUL disclosures)
- Analytical tools (NotebookLM)

Supporting documentation and validation records

Recommended Reading Flow

For a complete understanding of the report:

1. Read the main report (Section 1–4)
2. Use this appendix selectively:
 - Refer to **Phases** for process understanding
 - Refer to **Methodology** for model validation
 - Refer to **Evidence** for traceability



PART I — Development Phases & Process Log



Development Phases Timeline



1

**Vision & Research
Initiation (January 2026)**

2

**The Strategic Pivot
(February 2026)**

3

**Data Engineering &
Integrity (Technical Deep-
Dive)**

4

**Strategic Localization &
Critical Thinking (Strategy
Development)**

5

**Digital Architecture & UX
Design (March 2026)**

6

**Model Validation &
System Optimization
(Final Refinement Phase)**

Phase 1: Vision & Research Initiation (January 2026)

Initial Objective

Following the completion of the Data Analyst Associate certification, the project was initially conceived as a quantitative analysis and 2030 forecast of Hindustan Unilever Limited's (HUL) ESG performance.

Resource Discovery

Identified HUL's Digital Sustainability Reporting Center as a high-transparency data source, providing the raw material for an ESG-focused data portfolio.

Analytical Efficiency

To accelerate comprehension of multi-year reports, I curated a core dataset of official ESG filings and imported them into NotebookLM.

Using a **Targeted Query Strategy**—prompting based on specific data sources—I extracted high-value insights from dense text efficiently while maintaining accuracy.

The full query process is traceable via the NotebookLM link in the Evidence and Traceability section.



Phase 2: The Strategic Pivot (February 2026)

Environmental Scanning

A critical turning point occurred on February 9, following a report in eNanyang regarding the signing of 11 agreements and memoranda of understanding (MoUs) between Malaysia and India.

Insight Synthesis

This triggered a key realization: HUL's ESG maturity in India aligned with Malaysia's deepening institutional partnership with India.

By connecting geography, policy direction, and corporate capability, I identified a **"Policy Window"** for cross-border ESG transfer.

Strategic Re-positioning

Shifted the project from a corporate ESG audit to a **Cross-Border Green Strategy Blueprint**, centered on the transition from "Audit to Adaptation".

Phase 3: Data Engineering & Integrity (Technical Deep-Dive)

Data Quality Assurance (DQA)

Encountered data overlap and inconsistencies across HUL and Unilever global reports. Resolved this by narrowing the source scope to the 2023–2025 Digital Business Responsibility and Sustainability Reports (BRSR) to ensure audit-grade accuracy.

Mitigating AI Hallucination

Detected AI-generated filler data in Scope 3 due to missing supplier disclosures. Developed a custom **“Reporting Cycle” CSV** to enforce alignment with verified historical data.

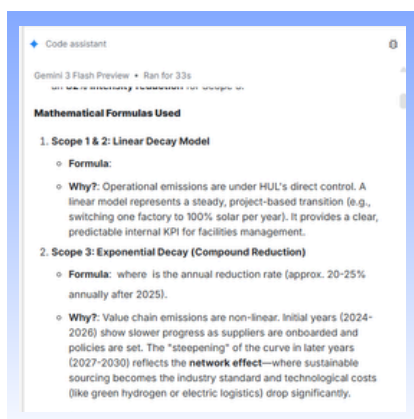
Predictive Modeling

Engineered a 2030 decarbonization forecast.

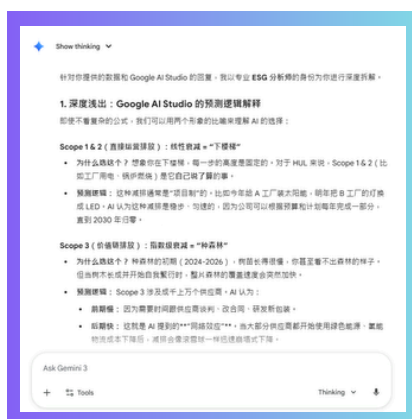
To validate model integrity:

- Extracted underlying formulas from Google AI Studio
- Conducted secondary audit using Google Gemini
- Benchmarked logic against SBTi guidelines

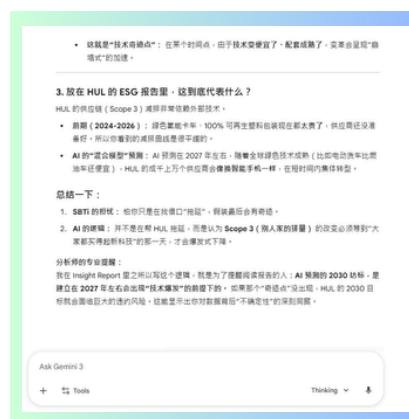
(Screenshots of validation process attached in Evidence section)



*This screenshot shows Google AI Studio explained the mathematical formulas and logic for section 3 data model



*This screenshot shows Google Gemini performing the secondary audit for section 3 data model



*This screenshot shows Google Gemini performing the secondary audit for section 3 data model

Phase 4: Strategic Localization & Critical Thinking (Strategy Development)

HUL "Shakti" & "Project Circular Bharat" Evaluation

I screened various HUL strategies for scalability, ultimately selecting the inclusive entrepreneurship model of Project Shakti and the systemic circularity framework of Project Circular Bharat.

Market Landscape Analysis

Identified key structural differences in Malaysia:

- **Urbanization:** dominance of modern retail (e.g., 99 Speedmart) vs India's rural Kirana networks
- **Platform Competition:** barriers from Shopee, TikTok vs India's ONDC ecosystem

Strategic Adaptation

Consequently, I Integrated Shakti and Circular Bharat, redesigning them into Circularity-as-a-Service Platform (CaaS Platform) — a "digital green service network" to better align with Malaysia's modern infrastructure and community needs.

Solution Optimization

During refinement of the CaaS Platform plan, identified **systemic synergy** across:

- CaaS Platform
- LC-SF
- Digital Plastic Recycling System
- RAP

This evolved into a **Circular ESG Ecosystem Strategy**, where each component reinforces the others.

AI was used to stress-test feasibility and ensure commercial viability.

Phase 5: Digital Architecture & UX Design (MARCH 2026)

Strategic Delivery Choice

I chose a Webpage format over a static document to:

- (1) enable Interactive Data Models for a professional user experience,
- (2) ensure the content is easily shareable while remaining harder to duplicate without authorization.

Tech Stack Optimization

Migrated development to Google AI Studio for higher efficiency and iterative speed in building the interactive web report.

UX/UI Innovation

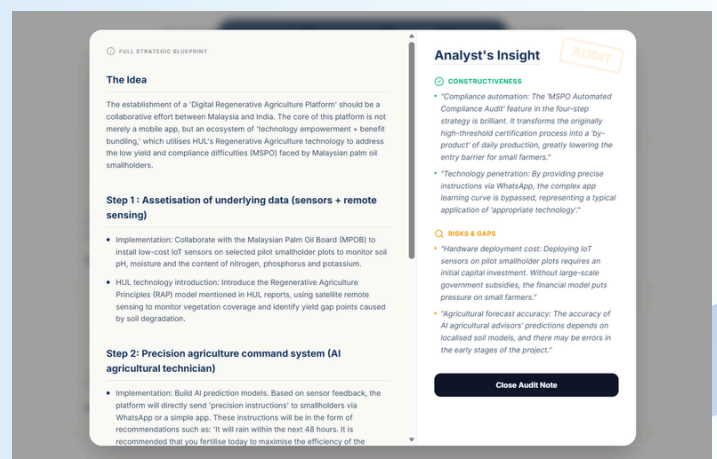
To manage six complex strategies in Section 3, I implemented a **Card-Based UI** to reduce cognitive load and enable modular exploration.

Content Preservation

During development in Google AI Studio, I observed that large data uploads caused model instability or "content swallowing." I resolved this via a **Segmented Upload Strategy**—converting sections into modular PDFs and building incrementally to preserve all details.



*This screenshot shows Interactive Data Models



*This screenshot shows Card-Based UI

Phase 6: Model Validation & System Optimization (Final Refinement Phase)

Model Integrity Check

During final validation of the interactive dashboard, I identified critical inconsistencies within Section 2.2B:

- Misaligned hover interactions across models
- Inconsistent relationships between Emission Trajectory and Reduction Contribution Unit ambiguity (MT vs Share %) leading to incorrect interpretation

Root Cause Diagnosis

Further investigation revealed that the issue was not isolated to visualization, but structural:

- Models were mathematically correct individually
- But lacked a fully aligned **causal linkage across the system**

Mathematical Reconciliation

Reconstructed the entire modeling framework to enforce:

- A single causal chain: **Adoption → Reduction Share → Emission Outcome**
- Full consistency across all three models
- Elimination of additive misinterpretation within a multiplicative system

System Standardization

Redefined Reduction Contribution as **Share (%)** instead of absolute MT to align with model structure and avoid false equivalence with emission outputs.

Structural Insight Expansion

Through debugging, a broader issue emerged:

- Report structure was overly dense
- Section transitions lacked clarity
- Key insights were not immediately visible

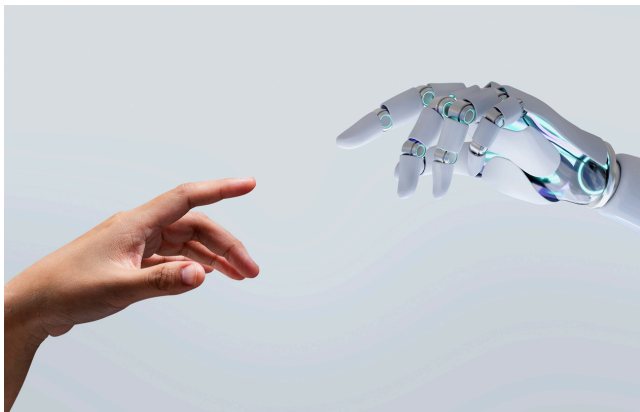
Report Optimization Phase

Initiated a full refinement of the report:

- Simplified Section 2 structure (2.2A → 2.2B → 2.2C flow)
- Strengthened Section transitions (especially Section 2 → 3)
- Introduced diagram-driven explanations
- Compressed interpretation layers for clarity and readability

This phase marked a transition from **model construction** → **system coherence** → **communication clarity**, ensuring that both analytical rigor and reader understanding were aligned.

Key Learnings & Reflections

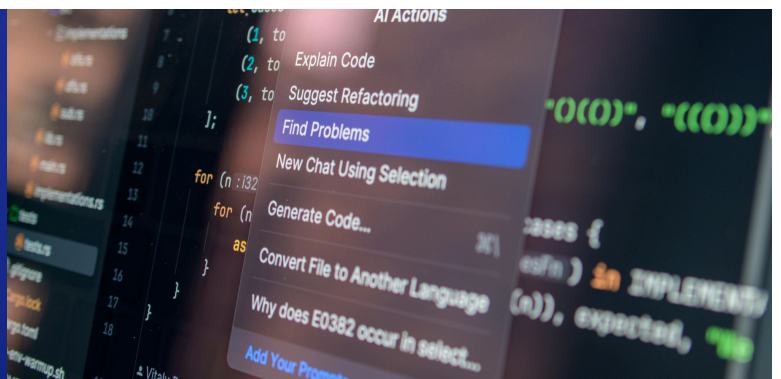


The "Human-in-the-Loop" Necessity

While AI significantly accelerated data synthesis (e.g., via NotebookLM), the strategic pivot originated from human judgment. Identifying the ESG “policy window” required contextual and geopolitical awareness beyond current AI capability.

Data Integrity as a Foundation

Resolving discrepancies between HUL and global reports demonstrated that data cleaning is a form of governance. Constraining AI inputs proved essential to achieving audit-grade outputs.



From Analyst to Strategist

The project evolved from interpreting historical data to designing forward-looking systems. The transition from Section 2 (diagnosis) to Section 3 (localization) reflects a shift toward system-level thinking.

Model Validity vs Communication Clarity

A technically correct model is insufficient if its logic is not intuitively understood. The final phase highlighted that clarity of structure is as critical as mathematical accuracy.



PART II — Model Methodology & Validation

2.2A Emission Reality Model Methodology and Structural Interpretation

This model is constructed using a descriptive-analytical approach based on reported emissions data across Scope 1, Scope 2, and Scope 3.

The methodology consists of two integrated layers:

1. Emission Trend Analysis (Quantitative Layer)

A time-series comparison is performed across all three emission scopes to evaluate:

- relative magnitude
- directional trends
- divergence in reduction patterns

This enables the identification of **relative differences in emission behavior** across scopes, and provides an empirical basis to assess where emission reductions appear more or less responsive to firm-level actions.

2. Structural Decomposition (Qualitative Layer)

Due to the absence of detailed category-level quantitative data, Scope 3 emissions are interpreted through a **qualitative decomposition approach**.

Key emission drivers are identified based on disclosed material categories, including:

- upstream procurement
- packaging systems
- logistics activities
- downstream product use

Rather than assigning proportional weights, this layer focuses on:

- identifying structural distribution
- mapping dependency across the value chain

No quantitative allocation is performed at this stage.

3. Integrated Interpretation

The model combines trend observation with structural mapping to assess:

- where emission reductions appear operationally feasible
- where systemic constraints are likely to dominate
- how value chain complexity influences decarbonization potential

This interpretation is **inferential**, based on observable patterns and disclosed structural drivers, rather than causal attribution.

Methodological Positioning

This approach prioritizes:

- structural accuracy over artificial precision
- interpretability over numerical completeness

It is designed to provide **decision-relevant insights under data constraints**, while avoiding unsupported quantitative assumptions.

Model Data Base

This model is based on real data disclosed in HUL's official reports.

Reported Emissions by Scope (Unit: tCO₂e)

- **2023** — Scope 1: 20,165 | Scope 2: 0 | Scope 3: 8,719,253
- **2024** — Scope 1: 14,622 | Scope 2: 96 | Scope 3: 9,962,535
- **2025** — Scope 1: 8,944 | Scope 2: 0 | Scope 3: Not disclosed

Note:

Scope 3 remains the dominant emission source, exceeding Scope 1 and 2 by several orders of magnitude.

Readers can refer to:

- HUL Sustainability Performance Data
- HUL Digital Business Responsibility and Sustainability Report (BRSR)

in the **Evidence and Traceability section** for source validation.

2.2B — Adoption Impact System Methodology

System Conceptual Foundation

The system is designed to capture Scope 3 decarbonization as a **system-level phenomenon**, rather than a firm-level optimization problem.

Unlike Scope 1 and Scope 2 emissions, which are largely controllable through internal operational improvements, Scope 3 emissions are distributed across a network of external stakeholders. As a result, emission reduction depends on the **rate and scale of supplier participation**.

To reflect this, the system integrates two interdependent dynamics:

- Internal efficiency improvements (firm-controlled)
- External adoption dynamics (ecosystem-driven)

These dynamics are **mathematically coupled**, ensuring that emission outcomes reflect real-world dependency between internal action and external participation.

System Integration

Causal Chain

The system operates as a sequential process:

$$A(t) \rightarrow \text{Reduction Driver} \rightarrow E(t)$$

Functional Roles

Component	Role
Adoption Curve	Defines participation level
Contribution Model	Explains driver dominance
Emission Model	Converts participation into outcomes

Consistency Principle

All outputs are derived from:

- the same adoption function
- the same emission structure

ensuring:

- internal consistency
 - interpretability
 - replicability
-

System Limitations

- Adoption impact coefficient is assumed
 - Supplier heterogeneity is not modeled
 - Policy and regulatory shocks are excluded
 - Time lags are simplified
-

System Key Insight

The system leads to a structural conclusion:

Scope 3 decarbonization cannot be achieved through internal efficiency alone.

It requires coordinated supplier participation, where adoption acts as the **primary driver of system-wide emission reduction**.

Adoption Curve Model — Methodology

Model Overview

The Adoption Curve models how supplier participation evolves over time and functions as the **primary driver of system activation**.

It is treated as an **exogenous input** to the system, determining the level of ecosystem engagement that enables emission reduction.

Mathematical Structure

$$A(t) = \frac{K}{1 + e^{-k(t-t_0)}}$$

$$A(t) = K / (1 + e^{-k(t - t_0)})$$

Component Interpretation

- **K** — Maximum adoption level (saturation point)
- **k** — Growth rate (speed of adoption diffusion)
- **t₀** — Inflection point (transition to rapid adoption phase)

Behavioral Characteristics

The function captures three distinct phases:

Initial Phase (Year 2025 - 2027)

Slow growth due to onboarding friction and coordination barriers

Acceleration Phase (Year 2028 - 2031)

Rapid adoption driven by network effects and ecosystem alignment

Saturation Phase (Year 2032 - 2035)

Diminishing growth as adoption approaches maximum capacity

Model Role

- Defines **ecosystem activation level**
- Acts as the **independent variable** driving all downstream effects
- Introduces **non-linearity** into the system

Methodological Positioning

The adoption curve is:

- A **structural representation**, not an empirical forecast
- Parameterized based on assumed system behavior
- Used to simulate how participation dynamics influence outcomes

Model Dataset in Main Report (Supplier Adoption Rate, %)

- 2025: 15%
- 2026: 20%
- 2027: 28%
- 2028: 38%
- 2029: 50%
- 2030: 60%
- 2031: 68%
- 2032: 74%
- 2033: 78%
- 2034: 81%
- 2035: 83%

Reduction Contribution Model — Methodology

Model Overview

The Reduction Contribution Model decomposes total emission reduction into:

- Internal efficiency contribution
- Adoption-driven contribution

It expresses contributions as **relative shares**, not absolute quantities.

Mathematical Structure

Step 1 — Total Reduction Definition

$$\Delta E(t) = E_{baseline}(t) - E_{modeled}(t)$$

$$\Delta E(t) = E_{baseline}(t) - E_{modeled}(t)$$

Step 2 — Contribution Decomposition

$$Share_{adoption}(t) = \frac{\alpha \cdot A(t)}{r_{base} + \alpha \cdot A(t)}$$

$$Share_{internal}(t) = 1 - Share_{adoption}(t)$$

$$Share_{adoption}(t) = (\alpha \cdot A(t)) / (r_{base} + \alpha \cdot A(t))$$

$$Share_{internal}(t) = 1 - Share_{adoption}(t)$$

Component Interpretation

1. Total Reduction ($\Delta E(t)$)

Represents the absolute difference between baseline and modeled emissions.

- Captures the overall decarbonization outcome
- Serves as the reference for contribution analysis
- Does not explain causality on its own

2. Internal Reduction Driver (r_{base})

Represents firm-controlled efficiency improvements.

- Modeled as a constant decay rate
- Reflects operational optimization (energy, process, etc.)
- Independent of external ecosystem participation

3. Adoption Driver ($A(t)$)

Represents the level of supplier participation over time.

- Modeled using a logistic (S-curve) function
- Captures network effects and scaling dynamics
- Acts as an external system-level variable

4. Adoption Impact Coefficient (α)

Represents the effectiveness of adoption in reducing emissions.

- Translates participation into actual emission impact
- Controls the strength of ecosystem influence
- Assumed parameter (not empirically calibrated)

5. Contribution Share (Key Interpretation Layer)

$$Share_{adoption}(t) = \frac{\alpha \cdot A(t)}{r_{base} + \alpha \cdot A(t)}$$

- Measures relative influence, not absolute reduction
- Ensures internal + adoption = 100%

Reflects how system dynamics evolve over time

Interpretation Logic

- The formulation provides a **normalized attribution framework**, designed to compare the relative influence of internal efficiency and adoption dynamics.
- It operates as an **interpretive layer**, rather than a direct mathematical decomposition of the emission function.
- The objective is to capture how **system-level influence evolves**, not to isolate exact reduction pathways.

Conceptual Meaning

The model is designed to **explain system behavior, not reconstruct physical outcomes**.

- It does not attempt to allocate precise emission reductions to individual drivers
- It does not convert contributions into absolute emission quantities

Instead, it reveals how the **dominant driver of decarbonization shifts over time**, as adoption scales within the ecosystem

Why Share (%) Instead of Absolute Values

- Ensures alignment with the **multiplicative structure** of the emission model
- Avoids **false precision** in attributing reductions
- Enhances clarity by expressing **relative system influence** rather than absolute outputs

Methodological Positioning

The model is positioned as:

- A **structural interpretation framework**, not a physical accounting model
- A **causal lens** to understand system dynamics
- A tool for analyzing **relative influence**, rather than measuring exact outcomes

Model Dataset in Main Report (Share of Total Reduction, %)

- 2025 — Adoption: 9% | Internal: 91%
- 2026 — Adoption: 12% | Internal: 88%
- 2027 — Adoption: 17% | Internal: 83%
- 2028 — Adoption: 23% | Internal: 77%
- 2029 — Adoption: 30% | Internal: 70%
- 2030 — Adoption: 36% | Internal: 64%
- 2031 — Adoption: 41% | Internal: 59%
- 2032 — Adoption: 44% | Internal: 56%
- 2033 — Adoption: 47% | Internal: 53%
- 2034 — Adoption: 49% | Internal: 51%
- 2035 — Adoption: 50% | Internal: 50%

Emission Trajectory Model — Methodology

Model Overview

The Emission Trajectory Model represents how emissions evolve under the combined influence of:

- Internal efficiency improvements
- Supplier adoption dynamics

Mathematical Structure

$$E_t = E_0 \cdot (1 - r_{base})^t \cdot (1 - \alpha \cdot A(t))$$

$$E_t = E_0 \cdot (1 - r_{base})^t \cdot (1 - \alpha \cdot A(t))$$

Component Interpretation:

1. Internal Reduction Component

$$(1 - r_{base})^t$$

- Represents firm-level efficiency improvements
- Modeled as exponential decay
- Reflects controllable internal actions

2. Adoption-Driven Component

$$(1 - \alpha \cdot A(t))$$

- Captures impact of supplier participation
- $A(t)$: adoption level
- α : effectiveness of adoption

Model Interpretation

The structure is **multiplicative**, not additive.

This implies:

- Internal efficiency alone cannot achieve deep decarbonization
- Adoption acts as a **scaling mechanism**
- Low adoption constrains system-wide impact

Methodological Positioning

The model is:

- Analytical (not predictive)
- Scenario-based (not empirical forecasting)
- Designed to illustrate structural dependency

Model Dataset in Main Report (Unit: MT)

- 2025 — Baseline: 10.0 | Modeled: 9.1
 - 2026 — Baseline: 10.1 | Modeled: 8.9
 - 2027 — Baseline: 10.2 | Modeled: 8.5
 - 2028 — Baseline: 10.3 | Modeled: 7.9
 - 2029 — Baseline: 10.4 | Modeled: 7.3
 - 2030 — Baseline: 10.5 | Modeled: 6.7
 - 2031 — Baseline: 10.6 | Modeled: 6.3
 - 2032 — Baseline: 10.7 | Modeled: 5.9
 - 2033 — Baseline: 10.8 | Modeled: 5.7
 - 2034 — Baseline: 10.9 | Modeled: 5.6
 - 2035 — Baseline: 11.0 | Modeled: 5.5
-

PART III — Evidence and Traceability

Evidence and Traceability

This section consolidates the core data sources, analytical tools, and reference materials underpinning the report, ensuring full transparency and enabling independent validation of the methodology and findings.

01 _____

Webpage report structure:



02 _____

Data Source

-  [HUL Sustainability Performance Data](#)
-  [HUL Digital Business Responsibility and Sustainability Reports \(BRSR\)](#)

03 _____

AI Tool



04 _____

News



Closing Statement

This appendix documents the full development process, methodological structure, and supporting evidence behind the report.

It serves not only as a record of how the work was constructed, but also as a framework that allows the analysis to be examined, validated, and replicated.

Together, these elements reflect a consistent approach:
from data interpretation → system modeling → strategic application.

The objective is not only to present conclusions, but to make the underlying logic transparent—so that the work can be understood, tested, and extended.

About the Author



THAM KE LI

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The author is a Strategic Consultant and Systems Architect specializing in Business Intelligence and ESG Strategy. This report marks a deliberate transition from technical observation to systemic solution architecture, integrating real-world data validation with forward-looking simulations into a unified transformation framework.

Moving beyond isolated metrics, the author constructs coherent, scalable systems that bridge the gap between high-level policy and operational execution. This work reflects a commitment to translating multi-dimensional chaos into clear logic and decision-ready strategy.

The author's methodology is informed by a diverse professional journey—spanning frontline sales, digital media, and independent philosophy. This background shapes a structured three-step cognitive process:

Receive (observe) ⇨ Process (structure) ⇨ Release (guide)
—balancing analytical rigor with human-centered strategic insight.

Driven by the pursuit of dynamic balance, the author focuses on how systems evolve and scale over time. Seeking environments that value cross-domain synthesis and integrity-driven strategy, the author aims to turn systemic challenges into actionable, high-impact solutions.